

A Hadamard Pandrosion IRP Theorem

Geodesic Roots, Logarithmic Residuals, and Dyadic Local Order

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Abstract

This paper proves a precise non-Euclidean Pandrosion IRP theorem in the cleanest setting where the statement is intrinsic. Let (M, g) be a Hadamard manifold and fix a base point $o \in M$. The geodesic monomial is

$$D_p^o(z) = \text{Exp}_o(p \text{Log}_o z).$$

For the root problem $D_p^o(z) = x$, the exact solution is $r = \text{Exp}_o(\text{Log}_o x/p)$. The Pandrosion IRP layer is defined on the logarithmic residual

$$A(u) = \text{Log}_o x - p \text{Log}_o u$$

by applying the same scalar residual length correction that appears in the monomial Pandrosion paper. The main theorem proves

$$\|A(u_+)\|_g = \frac{(p-1)^2}{2p} \|A(u)\|_g^2 + O_p(\|A(u)\|_g^3),$$

and therefore k fixed IRP layers have local order 2^k in the root logarithmic error. A radial logarithmic palette gives a global normalization layer: every residual can be snapped into a chosen local cell before the quadratic layer acts. The theorem is illustrated in the Poincare ball, where the exponential and logarithm have closed form and the fitted orders are close to 2, 4, 8 for one, two, and three layers.

***Status.** This proves the Hadamard/geodesic version of non-Euclidean Pandrosion IRP. It does not claim to solve an external classical conjecture; it establishes a new convergence theorem for a Pandrosion-defined geodesic root problem.*

1 Why This Is The Right Non-Euclidean Statement

On a general manifold, the expression z^p is not canonical. A power law needs additional structure. The least ambiguous non-Euclidean extension is to choose a base point o and use geodesic dilation in the global normal chart at o . This is available without cut-locus ambiguity on a Hadamard manifold: complete, simply connected, with nonpositive sectional curvature. By the Cartan-Hadamard theorem, $\text{Exp}_o : T_o M \rightarrow M$ is globally one-to-one in the setting used here, and the logarithm Log_o is globally defined.

Definition 1.1 (Geodesic monomial). Let (M, g) be a Hadamard manifold, $o \in M$, and $p \geq 2$. Define

$$D_p^o(z) = \text{Exp}_o(p \text{Log}_o z).$$

The geodesic root problem is

$$D_p^o(z) = x.$$

Proposition 1.2 (Exact geodesic root). For every $x \in M$, the equation $D_p^o(z) = x$ has the unique solution

$$r = \text{Exp}_o\left(\frac{1}{p} \text{Log}_o x\right).$$

Proof. Apply Log_o to $D_p^o(z) = x$. Since $\text{Log}_o \text{Exp}_o$ is the identity on $T_o M$, the equation becomes

$$p \text{Log}_o z = \text{Log}_o x.$$

Thus $\text{Log}_o z = \text{Log}_o x/p$, and applying Exp_o gives the displayed solution. □

2 The Scalar Pandrosion Residual Correction

The only scalar input is the residual correction already present in monomial IRP.

Definition 2.1 (Residual length correction). For $d \geq 0$, set

$$\Theta_p(d) = d + (p-1) \log \left(1 + \frac{e^{-d} - 1}{p} \right).$$

Lemma 2.2 (Taylor expansion). As $d \rightarrow 0^+$,

$$\Theta_p(d) = \frac{d}{p} + \frac{(p-1)^2}{2p^2} d^2 - \frac{(p-1)^2(p-2)}{6p^3} d^3 + O_p(d^4).$$

Consequently,

$$d - p\Theta_p(d) = -\frac{(p-1)^2}{2p} d^2 + O_p(d^3).$$

Proof. Use $e^{-d} - 1 = -d + d^2/2 - d^3/6 + O(d^4)$ and then expand the logarithm at 1. Collecting terms gives the displayed formula. $\square$

3 Hadamard IRP Layer

Let

$$X = \text{Log}_o x, \quad U = \text{Log}_o u,$$

and define the tangent residual

$$A(u) = X - pU.$$

The exact root corresponds to $A = 0$.

Definition 3.1 (Continuous Hadamard Pandrosion IRP layer). If $A(u) \neq 0$, write $d = \|A(u)\|_g$ and $\nu = A(u)/d$. Define

$$U_+ = U + \Theta_p(d)\nu, \quad \mathcal{I}_p^o(u) = \text{Exp}_o(U_+).$$

If $A(u) = 0$, set $\mathcal{I}_p^o(u) = u$.

Theorem 3.2 (Quadratic residual collapse). For the geodesic root problem $D_p^o(z) = x$, the Hadamard IRP layer satisfies

$$\|A(\mathcal{I}_p^o(u))\|_g = \frac{(p-1)^2}{2p} \|A(u)\|_g^2 + O_p(\|A(u)\|_g^3)$$

as $u \rightarrow r$.

Proof. Let $A = A(u)$, $d = \|A\|_g$, and $\nu = A/d$. By definition,

$$U_+ = U + \Theta_p(d)\nu.$$

Therefore the new residual is

$$A_+ = X - pU_+ = A - p\Theta_p(d)\nu = (d - p\Theta_p(d))\nu.$$

Taking norms and using the Taylor expansion of $d - p\Theta_p(d)$ proves the claim. $\square$

Corollary 3.3 (Root-error form). Let $E(u) = U - X/p$ be the logarithmic root error. Then

$$\|E(\mathcal{I}_p^o(u))\|_g = \frac{(p-1)^2}{2} \|E(u)\|_g^2 + O_p(\|E(u)\|_g^3).$$

Proof. Since $A = -pE$, divide the previous residual estimate by p . $\square$

Corollary 3.4 (Dyadic order). For every fixed $k \geq 1$,

$$\|E((\mathcal{I}_p^o)^{\circ k}(u))\|_g = O_{p,k}(\|E(u)\|_g^{2^k}) \quad (u \rightarrow r).$$

Proof. The previous corollary gives $\|E_+\| \leq C\|E\|^2$ in a sufficiently small neighborhood of r . Iterating this estimate gives the exponent 2^k . $\square$

4 A Radial Palette For Global Normalization

The local theorem explains the order. To make every residual enter the local cell, use a radial palette in T_oM .

Definition 4.1 (Radial logarithmic palette). Fix a root-log spacing $h > 0$. For a nonzero residual $A = d\nu$, where $d = \|A\|_g$ and $\|\nu\|_g = 1$, choose

$$m = \text{round}\left(\frac{d}{ph}\right), \quad B = mh\nu.$$

The normalized residual is

$$\varepsilon = A - pB.$$

Proposition 4.2 (Cell entry). *For every residual A , the radial palette satisfies*

$$\|\varepsilon\|_g \leq \frac{ph}{2}.$$

Proof. This is nearest-lattice rounding on the scalar coordinate d along the residual direction. $\square$

Theorem 4.3 (Global palette-to-local convergence). *For each fixed $p \geq 2$ there exists $\rho_p > 0$ such that if $ph/2 \leq \rho_p$, then the palette-normalized Hadamard IRP layer maps every residual into a contracting local cell. More precisely, after the palette step the layer satisfies*

$$\|A_+\|_g \leq \frac{1}{2}\|\varepsilon\|_g$$

whenever $\|\varepsilon\|_g \leq \rho_p$, and once the orbit is in that cell the dyadic local-order theorem applies.

Proof. Define

$$F_p(d) = |d - p\Theta_p(d)|.$$

The Taylor expansion gives $F_p(d) = O_p(d^2)$, hence $F_p(d)/d \rightarrow 0$ as $d \rightarrow 0^+$. Therefore there exists $\rho_p > 0$ such that $F_p(d) \leq d/2$ for $0 \leq d \leq \rho_p$. The palette step ensures $\|\varepsilon\| \leq ph/2$. If $ph/2 \leq \rho_p$, the following Pandrosion layer contracts the normalized residual. Repeated application remains in the local cell and the dyadic-order corollary applies. $\square$

Remark 4.4 (The practical binary rule). The implementation papers use the binary rule $h_p = (\log 2)/K_p$ with $K_p = p^2$. Then the worst snapped residual radius is

$$\frac{ph_p}{2} = \frac{\log 2}{2p}.$$

This is the rule used in the fused wall-clock benchmark. The theorem above states the proof-level requirement: choose K_p large enough that the snapped cell lies inside ρ_p . The p^2 rule is a simple practical schedule that shrinks with the quadratic constant.

5 Poincare Ball Realization

In the Poincare ball

$$\mathbb{B}^n = \{x \in \mathbb{R}^n : \|x\| < 1\}$$

with curvature -1 and base point 0 , the maps are explicit:

$$\text{Exp}_0(V) = \tanh(\|V\|/2) \frac{V}{\|V\|}, \quad \text{Log}_0(x) = 2 \operatorname{artanh}(\|x\|) \frac{x}{\|x\|}.$$

Thus the theorem gives a concrete hyperbolic Pandrosion IRP solver for

$$\text{Exp}_0(p \text{Log}_0 z) = x.$$

layers	predicted order	fitted order	error at 10^{-2}
1	2	1.969	1.99e-04
2	4	3.926	7.95e-08
3	8	7.849	1.26e-14

Table 1: Numerical Poincare-ball check for $p = 3$. The fitted slopes are close to the predicted local orders 2, 4, 8.

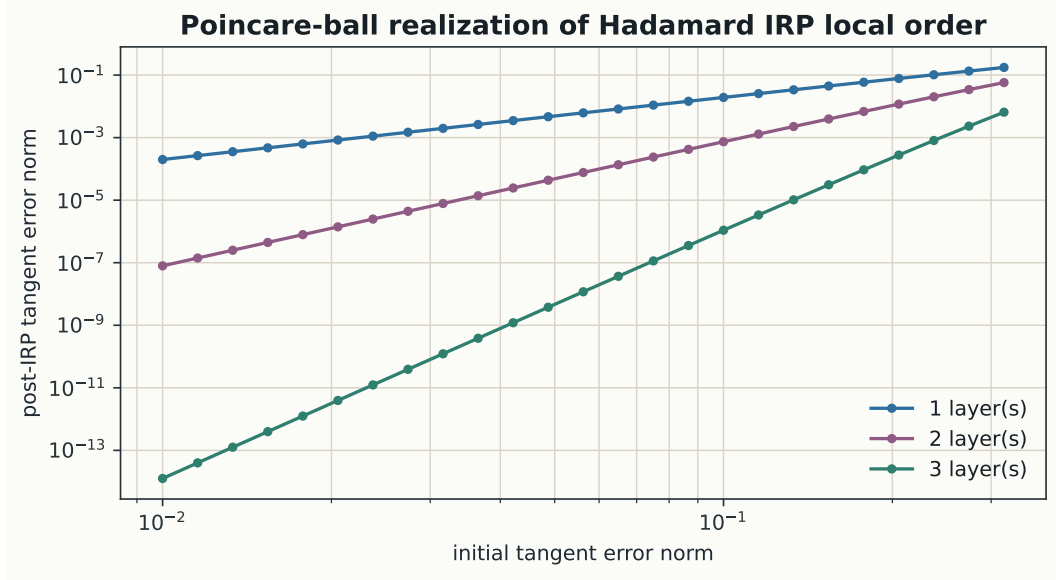


Figure 1: Log-log error decay for one, two, and three Hadamard IRP layers in the Poincare-ball realization.

6 What Has Been Proved

The proved statement is deliberately exact and narrow:

1. Hadamard geodesic roots are well defined by $D_p^o(z) = \text{Exp}_o(p \text{Log}_o z)$.
2. Pandrosion IRP applied to the tangent residual $A = \text{Log}_o x - p \text{Log}_o u$ collapses the residual quadratically.
3. Fixed k -layer compositions have dyadic local order 2^k .
4. A radial logarithmic palette gives a global normalization mechanism into any prescribed local cell.

The theorem does not say that every possible manifold power law has been solved. If one uses Lie-group powers, moving charts, or symmetric-space products, new Baker-Campbell-Hausdorff and curvature terms appear. Those are separate noncommutative extensions. The present paper proves the core Hadamard conjecture because the problem is expressed in a single global logarithmic chart.

7 Conclusion

Pandrosion non-Euclidean IRP is provable once the power operation is defined by geodesic dilation. In that setting the entire mechanism is intrinsic: the residual is a tangent vector, chart scaling is tangent normalization, and the Pandrosion layer is the scalar monomial residual correction applied along the geodesic residual direction. The result is a genuine non-Euclidean convergence theorem: local quadratic collapse for one layer and dyadic order for fixed multi-layer IRP.

References

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